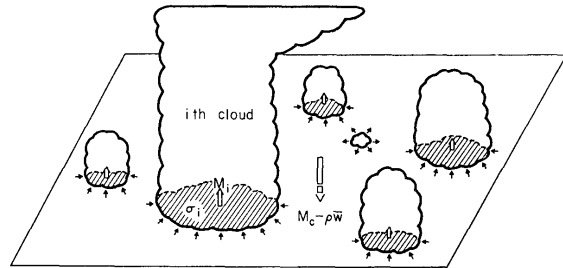


Interaction of a Cumulus Cloud Ensemble with the Large-Scale Environment, Part I



Collective effect of clouds

Rising mass flux, compensating subsidence and detrainment change environmental heat and moisture.

$$M_c = \sum_i \rho \sigma_i w_i$$

σ_i : area fraction.
 w_i : vertical velocity.

The theory couples a cumulus population to its large-scale environment

Arakawa & Schubert (1974): the cloud ensemble

Arakawa and Schubert develop a theory for the mutual interaction of a cumulus ensemble and its slowly evolving large-scale environment, including the subcloud mixed layer. The aim is a cumulus parameterization for models that cannot resolve individual clouds.

Figure 1 is the original conceptual sketch of clouds in a representative horizontal area. It illustrates mass exchange and is not an observed map or a statistical model for cloud locations. Each cloud carries the mass flux $M_i = \rho \sigma_i w_i$, where σ_i is its area fraction. The sum gives the ensemble mass flux.

The environmental budgets contain warming and drying by compensating subsidence, and cooling and moistening by detrained air and evaporation of detrained water. The theory reduces the collective effect to the total cloud mass flux, detrainment and detrained liquid water. It assumes a small active cloud fraction and a horizontally uniform environment. Source: Sections 2–3, Figure 1, Equations (7), (74)–(75).

A spectrum of entraining plumes

Each cloud type has a fractional entrainment rate λ .

More entrainment dilutes the plume.
In this sounding, its top is lower.

The remaining unknown is $\mathcal{M}_B(\lambda)$:
the mass flux allocated to each type at the mixed-layer top.

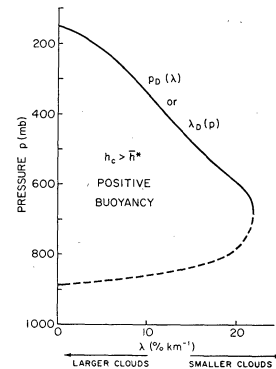


Fig. 8: cloud-top pressure vs. entrainment.

The environment fixes the vertical structure of each cloud type

Entrainment sets the cloud type and cloud top

The ensemble is split into cloud types labelled by the fractional entrainment rate λ , assumed constant with height. Each type is represented by an entraining plume rooted in the mixed layer. The top is where buoyancy vanishes. Given the environmental profiles and mixed-layer properties, the model gives the thermodynamic structure and detrainment level of each type.

Figure 8 shows cloud-top pressure against entrainment for the tropical sounding used in the paper. Focus on the solid cloud-top curve: larger entrainment mixes in more environmental air and gives a lower top in this example, so the top pressure increases.

What remains unknown is $\mathcal{M}_B(\lambda)$, the mass-flux density over cloud types at the mixed-layer top. Integrating it over a range of λ gives the flux assigned to that range. It is not a cloud-number density. In the related tower model, entrainment is inversely related to radius, but no observed radius distribution is supplied. Source: Sections 4-5, Figure 8 and Equation (94).

Quasi-equilibrium closes the mass-flux spectrum

The cloud work function $A(\lambda)$ is an integral of cloud buoyancy, weighted by the normalized plume mass flux. It measures kinetic-energy generation per unit mass flux at the mixed-layer top. Its tendency separates into a cloud contribution, linear in $\mathcal{M}_B$ through the kernel K , and the large-scale forcing F .

The proposed closure assumes that clouds adjust on a shorter timescale than the forcing changes. The paper estimates an adjustment time of 10^3 – 10^4 s against a large-scale timescale of at least 10^5 s. To leading order, cloud stabilization balances forcing. This gives the integral equation for the nonnegative mass-flux spectrum. The equality applies to active types, with an inequality for absent types. Quasi-equilibrium does not mean steady temperature or moisture.

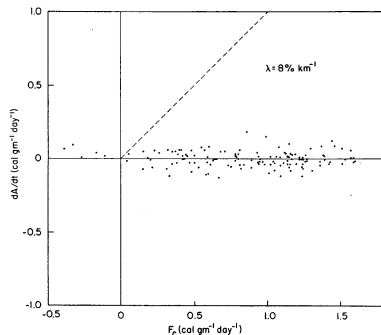
The original upper panel of Figure 13 compares the observed tendency of A with cloud-layer forcing F_C for $\lambda = 8\% \text{ km}^{-1}$ in the Marshall Islands. The dashed line is $dA/dt = F_C$, the response without cloud adjustment. The observations lie much nearer zero, supporting the closure in this setting. Mixed-layer forcing was not computed from these data. Part I derives the closure and tests its premise, while the numerical solution method and applications are deferred to Part II. Source: Sections 6–7, Equation (150), Figure 13 and the end of Section 8.

Cloud work function $A(\lambda)$

An integral measure of plume buoyancy. Clouds adjust faster than the large-scale forcing changes.

$$\int K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) \simeq 0$$

Cloud stabilization balances the forcing. This equation constrains $\mathcal{M}_B(\lambda)$ for active cloud types.



Marshall Islands: observed dA/dt is small relative to the forcing.

Quasi-equilibrium provides the closure for the ensemble's mass flux

No distributions reported for

Thermal size at a specified height.

Thermal positions in the (x, y) plane.

Distances between thermals.

The closure determines mass flux. Cloud number and lifetime remain separate unknowns.

Possible use: our inference

If the flux closure is solved, Eq. (145) gives a conditional route to cloud counts:

$$\mathcal{N}(\lambda) = \frac{\mathcal{M}_B(\lambda) \tau(\lambda)}{m_B(\lambda)}$$

Needs lifetime τ , mass transported per cloud m_B , and then a cloud–thermal mapping.

Even a known cloud count would leave thermal positions and spacing open

What is missing for a 2D thermal field

The paper does not report distributions of thermal sizes, thermal coordinates in a horizontal plane, or distances between thermals. It provides an ensemble flux closure without a horizontal point process. The cloud sketch on slide 1 should not be mistaken for spatial evidence.

Equation (145) explicitly writes the flux density as $\mathcal{M}_B(\lambda) = \mathcal{N}(\lambda) m_B(\lambda) / \tau(\lambda)$, where $\mathcal{N}$ describes the cloud population, τ is lifetime, and m_B is the mass transported by a single cloud at the reference level over its whole lifetime, in the paper’s area convention. The rearrangement on the slide is algebra, not a newly measured distribution. The paper does not determine these factors separately.

Our proposed use is conditional: first solve the mass-flux closure in an applicable regime, then add independent cloud lifetime and single-cloud transport estimates to infer cloud counts by type. A calibrated relationship between cloud type and the size of its feeding updraft, plus a cloud-to-thermal association, would still be needed to say anything about thermal counts or sizes. Dry thermals would remain outside that cloud-selected population.

The tower-model relation between entrainment and radius, discussed around Equation (94), could supply an additional modelling assumption for a radius proxy. It cannot turn a mass-flux-weighted spectrum directly into a number-weighted thermal-size density. Neither the inferred count nor such a size proxy specifies clustering or nearest-neighbour distances. Choosing a Poisson point process, for example, would be a new untested assumption. The distinction between integrated budgets and spatial organization is the main limit for our purpose. Source: Sections 4 and 7, Equations (94) and (145).